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# GP-VAE: Deep Probabilistic Multivariate Time Series Imputation

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## Abstract

Multivariate time series with missing values are common in areas such as healthcare and finance, and have grown in number and complexity over the years. This raises the question whether deep learning methodologies can outperform classical data imputation methods in this domain. However, naïve applications of deep learning fall short in giving reliable confidence estimates and lack interpretability. We propose a new deep sequential latent variable model for dimensionality reduction and data imputation. Our modeling assumption is simple and interpretable: the high dimensional time series has a lower-dimensional representation which evolves smoothly in time according to a Gaussian process. The non-linear dimensionality reduction in the presence of missing data is achieved using a VAE approach with a novel structured variational approximation. We demonstrate that our approach outperforms several classical and deep learning-based data imputation methods on high-dimensional data from the domains of computer vision and healthcare, while additionally improving the smoothness of the imputations and providing interpretable uncertainty estimates.

## 1 Introduction

Time series are often associated with missing values, for instance due to faulty measurement devices, partially observed states, or costly measurement procedures [Honaker and King, 2010]. These missing values impair the usefulness and interpretability of the data, leading to the problem of *data imputation*: estimating

those missing values from the observed ones [Rubin, 1976].

Multivariate time series, consisting of multiple correlated univariate time series or *channels*, give rise to two distinct ways of imputing missing information: (1) by exploiting temporal correlations within each channel, and (2) by exploiting correlations across channels, for example by using lower-dimensional representations of the data. For instance in a medical setting, if the blood pressure of a patient is unobserved, it can be informative that the heart rate at the current time is higher than normal and that the blood pressure was also elevated an hour ago. An ideal imputation model for multivariate time series should therefore take both of these sources of information into account. Another desirable property of such models is to offer a probabilistic interpretation, allowing for uncertainty estimation.

Unfortunately, current imputation approaches fall short with respect to at least one of these desiderata. While there are many time-tested statistical methods for multivariate time series analysis (e.g., Gaussian processes [Roberts et al., 2013]) that work well in the case of complete data, these methods are generally not applicable when features are missing. On the other hand, classical methods for time series imputation often do not take the potentially complex interactions between the different channels into account [Little and Rubin, 2002, Pedersen et al., 2017]. Finally, recent work has explored the use of non-linear dimensionality reduction using variational autoencoders for i.i.d. data points with missing values [Nazabal et al., 2018, Ainsworth et al., 2018, Ma et al., 2018], but this work has not considered temporal data and strategies for sharing statistical strength across time.

Following these considerations, it is promising to combine non-linear dimensionality reduction with an expressive time series model. This can be done by jointly learning a mapping from the data space (where features are missing) into a latent space (where all dimensions

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are fully determined). A statistical model of choice can then be applied in this latent space to model temporal dynamics. If the dynamics model and the mapping for dimensionality reduction are both differentiable, the approach can be trained end-to-end.

In this paper, we propose an architecture that uses deep variational autoencoders (VAEs) to map the missing data time series into a latent space without missingness, where we model the low-dimensional dynamics with a Gaussian process (GP). As we will discuss below, we hereby propose a prior model that efficiently operates at multiple time scales, taking into account that the multivariate time series may have different channels (e.g., heart rate, blood pressure, etc.) that change with different characteristic frequencies. Finally, our variational inference approach makes use of efficient structured variational approximations, where we fit another multivariate Gaussian process in order to approximate the intractable true posterior.

We make the following contributions:

- **A new model.** We propose a VAE architecture for multivariate time series imputation with a GP prior in the latent space to capture temporal dynamics. We propose a Cauchy kernel to allow the time series to display dynamics at multiple scales in a reduced dimensionality.
- **Efficient inference.** We use a structured variational approximation that models posterior correlations in the time domain. By construction, inference is efficient and the time complexity for sampling from the variational distribution, used for training, is linear in the number of time steps (as opposed to cubic when done naïvely).
- **Benchmarking on real-world data.** We carry out extensive comparisons to classical imputation methods as well as state-of-the-art deep learning approaches, and perform experiments on data from two different domains. Our method shows favorable performance in both cases.

We start by reviewing the related literature in Sec. 2, describe the general setting in Sec. 3.1 and introduce our model and inference scheme in Sec. 3.2 and Sec. 3.3, respectively. Experiments and conclusions are presented in Sec. 4 and 5.

## 2 Related work

**Classical statistical approaches.** The problem of missing values has been a long-standing challenge in many time series applications, especially in the field of medicine [Pedersen et al., 2017]. The earliest approaches to deal with this problem often relied on

heuristics, such as mean imputation or forward imputation. Despite their simplicity, these methods are still widely applied today due to their efficiency and interpretability [Honaker and King, 2010]. Orthogonal to these ideas, methods along the lines of expectation-maximization (EM) have been proposed, but they often require additional modeling assumptions [Bashir and Wei, 2018].

**Bayesian methods.** When it comes to estimating likelihoods and uncertainties relating to the imputations, Bayesian methods, such as Gaussian processes (GPs) [Rasmussen, 2003], have a clear advantage over non-Bayesian methods such as single imputation [Little and Rubin, 2002]. There has been much recent work in making these methods more expressive and incorporating prior knowledge from the domain (e.g., medical time series) [Wilson et al., 2016, Fortuin and Rätsch, 2019] or adapting them to work on discrete domains [Fortuin et al., 2018a], but their wide-spread adoption is hindered by their limited scalability and the challenges in designing kernels that are robust to missing values. Our latent GP prior bears certain similarities to the GP latent variable model (GP-LVM) [Lawrence, 2004, Titsias and Lawrence, 2010], but in contrast to this line of work, we propose an efficient amortized inference scheme.

**Deep learning techniques.** Another avenue of research in this area uses deep learning techniques, such as variational autoencoders (VAEs) [Nazabal et al., 2018, Ainsworth et al., 2018, Ma et al., 2018, Dalca et al., 2019] or generative adversarial networks (GANs) [Yoon et al., 2018, Li et al., 2019]. It should be noted that VAEs allow for tractable likelihoods, while GANs generally do not and have to rely on additional optimization processes to find latent representations of a given input [Lipton and Tripathi, 2017]. Unfortunately, none of these models explicitly take the temporal dynamics of time series data into account. Conversely, there are deep probabilistic models for time series [e.g., Krishnan et al., 2015, 2017, Fortuin et al., 2018b], but those do not explicitly handle missing data. There are also some VAE-based imputation methods that are designed for a setting where the data is complete at training time and the missingness only occurs at test time [Garnelo et al., 2018a,b, Ivanov et al., 2018]. We do not regard this setting in our work.

**HI-VAE.** Our approach borrows some ideas from the HI-VAE [Nazabal et al., 2018]. This model deals with missing data by defining an ELBO whose reconstruction error term only sums over the observed part of the data. For inference, the incomplete data are filled with arbitrary values (e.g., zeros) before they are fed into the inference network, which induces an unavoidable bias. The main difference to our approach is that the

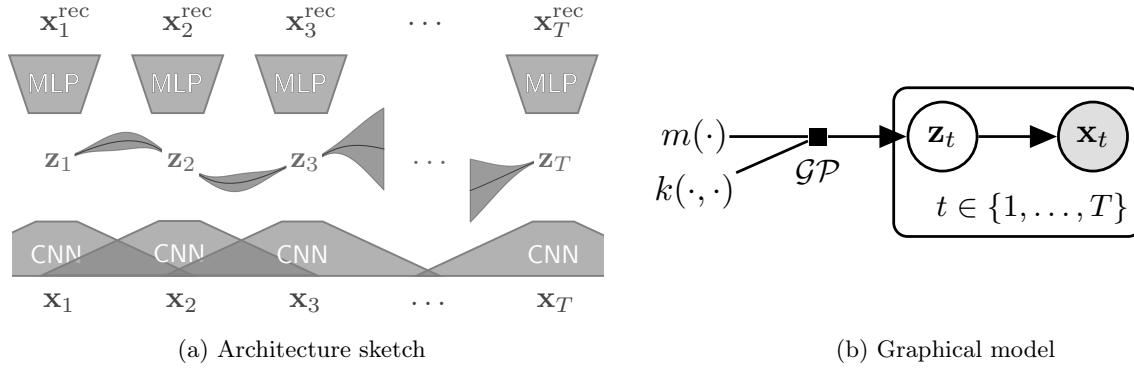


Figure 1: Overview of our proposed model with a convolutional inference network, a deep feed-forward generative network and a Gaussian process prior with mean function  $m(\cdot)$  and kernel function  $k(\cdot, \cdot)$  in latent space. The several CNN blocks as well as the MLP blocks are each sharing their respective parameters.

HI-VAE was not formulated for sequential data and therefore does not exploit temporal information in the imputation task.

### Deep learning for time series imputation.

While the mentioned deep learning approaches are very promising, most of them do not take the time series nature of the data directly into account, that is, they do not model the temporal dynamics of the data when dealing with missing values. To the best of our knowledge, the only deep generative model for missing value imputation that does account for the time series nature of the data is the GRUI-GAN [Luo et al., 2018], which we describe in Sec. 4. Another deep learning model for time series imputation is BRITS [Cao et al., 2018], which uses recurrent neural networks (RNNs). It is trained in a self-supervised way, predicting the observations in a time series sequentially. We compare against both of these models in our experiments.

**Other related work.** Our proposed model combines several ideas from the domains of Bayesian deep learning and classical probabilistic modeling; thus, removing elements from our model naturally relates to other approaches. For example, removing the latent GP for modeling dynamics as well as our proposed structured variational distribution results in the HI-VAE [Nazabal et al., 2018] described above. Furthermore, our idea of using a latent GP in the context of a deep generative model bears similarities to the GPPVAE [Casale et al., 2018] and can be seen as an extension of this model. While the GPPVAE allows for a joint GP prior over the whole data set through the use of a specialized inference mechanism, our model puts a separate GP prior on every time series and can hence rely on more standard inference techniques. However, our model takes missingness into account and uses a structured variational distribution, while the GPPVAE uses mean field inference and is designed for fully observed data.

Lastly, the GP prior with the Cauchy kernel is reminiscent of Jähnichen et al. [2018] and the structured variational distribution is similar to the one used by Bamler and Mandt [2017b] in the context of modeling word embeddings over time, neither of which used amortized inference.

## 3 Model

We propose a novel architecture for missing value imputation, an overview of which is depicted in Figure 1. Our model can be seen as a way to perform amortized approximate inference on a latent Gaussian process model.

The main idea of our proposed approach is to embed the data into a latent space of reduced dimensionality, in which every dimension is fully determined, and then model the temporal dynamics in this latent space. Since many features in the data might be correlated, the latent representation captures these correlations and uses them to reconstruct the missing values. Moreover, the GP prior in the latent space encourages the model to embed the data into a representation in which the temporal dynamics are smoother and more easily explainable than in the original data space. Finally, the structured variational distribution of the inference network allows the model to integrate temporal information into the representations, such that the reconstructions of missing values cannot only be informed by correlated observed features at the same time point, but also by the rest of time series.

Specifically, we combine ideas from VAEs [Kingma and Welling, 2014], GPs [Rasmussen, 2003], Cauchy kernels [Jähnichen et al., 2018], structured variational distributions with efficient inference [Bamler and Mandt, 2017b], and a special ELBO for missing data [Nazabal et al., 2018] and synthesize these ideas into a general

framework for missing data imputation on time series. In the following, we will outline the problem setting, describe the assumed generative model, and derive our proposed inference scheme.

### 3.1 Problem setting and notation

We assume a data set  $\mathbf{X} \in \mathbb{R}^{T \times d}$  with  $T$  data points  $\mathbf{x}_t = [x_{t1}, \dots, x_{tj}, \dots, x_{td}]^\top \in \mathbb{R}^d$ . Let us assume that the  $T$  data points were measured at  $T$  consecutive time points  $\tau = [\tau_1, \dots, \tau_T]^\top$  with  $\tau_t < \tau_{t+1} \forall t$ . By convention, we usually set  $\tau_1 = 0$ . The data  $\mathbf{X}$  can thus be viewed as a time series of length  $\tau_T$  in time.

We moreover assume that any number of these data features  $x_{tj}$  can be missing, that is, that their values can be unknown. We can now partition each data point into observed and unobserved features. The observed features of data point  $\mathbf{x}_t$  are  $\mathbf{x}_t^o := [x_{tj} | x_{tj} \text{ is observed}]$ . Equivalently, the missing features are  $\mathbf{x}_t^m := [x_{tj} | x_{tj} \text{ is missing}]$  with  $\mathbf{x}_t^o \cup \mathbf{x}_t^m \equiv \mathbf{x}_t$ .

We can now use this partitioning to define the problem of missing value imputation. Missing value imputation describes the problem of estimating the true values of the missing features  $\mathbf{X}^m := [\mathbf{x}_t^m]_{1:T}$  given the observed features  $\mathbf{X}^o := [\mathbf{x}_t^o]_{1:T}$ . Many methods assume the different data points to be independent, in which case the inference problem reduces to  $T$  separate problems of estimating  $p(\mathbf{x}_t^m | \mathbf{x}_t^o)$ . In the time series setting, this independence assumption is not satisfied, which leads to the more complex estimation problem of  $p(\mathbf{x}_t^m | \mathbf{x}_{1:T}^o)$ .

### 3.2 Generative model

In this subsection, we describe the details of our proposed approach of reducing the observed time series with missing data into a representation without missingness, and modeling dynamics in this lower-dimensional representation using Gaussian processes. Yet, it is tempting to try to skip the step of dimensionality reduction and instead directly try to model the incomplete data in the observed space using GPs. We argue that this is not practical for several reasons.

Gaussian processes are well suited for time series modeling [Roberts et al., 2013] and offer many advantages, such as data-efficiency and calibrated posterior probabilities. However, they come at the cost of inverting the kernel matrix, which has a time complexity of  $\mathcal{O}(n^3)$ . Moreover, designing a kernel function that accurately captures correlations in feature space and also in the temporal dimension is difficult.

This problem becomes even worse if certain observations are missing. One option is to fill the missing values with some numerical value (e.g., zero) to make the kernel computable. However, this arbitrary filling

may make two data points with different missingness patterns look very dissimilar when in fact they are close to each other in the ground-truth space. Another alternative is to treat every channel of the multivariate time series separately and let the GP infer missing values, but this ignores valuable correlations across channels.

In this work, we overcome the problem of defining a suitable GP kernel in the data space with missing observations by instead applying the GP in the latent space of a variational autoencoder where the encoded feature representations are complete. That is, we assign a latent variable  $\mathbf{z}_t \in \mathbb{R}^k$  for every  $\mathbf{x}_t$ , and model temporal correlations in this reduced representation using a GP,  $\mathbf{z}(\tau) \sim \mathcal{GP}(m_z(\cdot), k_z(\cdot, \cdot))$ . This way, we decouple the step of filling in missing values and capturing instantaneous correlations between the different feature dimensions from modeling dynamical aspects. The graphical model is depicted in Figure 1b.

A remaining practical difficulty that we encountered is that many multivariate time series display dynamics at multiple time scales. One of our main motivations is to model time series that arise in medical setups where doctors measure different patient variables and vital signs, such as heart rate, blood pressure, etc. When using conventional GP kernels (e.g., the RBF kernel,  $k_{RBF}(r | \lambda) = \exp(-\lambda r^2/2)$ ), one implicitly assumes a single time scale of relevance ( $\lambda$ ). We found that this choice does not reflect the dynamics of medical data very well.

In order to model data that varies at multiple time scales, we consider a mixture of RBF kernels with different  $\lambda$ 's [Rasmussen, 2003]. By defining a Gamma distribution over the length scale, that is,  $p(\lambda | \alpha, \beta) \propto \lambda^{\alpha-1} \exp(-\alpha\lambda/\beta)$ , we can compute an infinite mixture of RBF kernels,

$$\int p(\lambda | \alpha, \beta) k_{RBF}(r | \lambda) d\lambda \propto \left(1 + \frac{r^2}{2\alpha\beta^{-1}}\right)^{-\alpha}.$$

This yields the so-called Rational Quadratic kernel [Rasmussen, 2003]. For  $\alpha = 1$  and  $l^2 = 2\beta^{-1}$ , it reduces to the Cauchy kernel

$$k_{Cauchy}(\tau, \tau') = \sigma^2 \left(1 + \frac{(\tau - \tau')^2}{l^2}\right)^{-1}, \quad (1)$$

which has previously been successfully used in the context of robust dynamic topic modeling where similar multi-scale time dynamics occur [Jähnichen et al., 2018]. We therefore choose this kernel for our Gaussian process prior.

Given the latent time series  $\mathbf{z}_{1:T}$ , the observations  $\mathbf{x}_t$  are generated time-point-wise by

$$p_\theta(\mathbf{x}_t | \mathbf{z}_t) = \mathcal{N}(g_\theta(\mathbf{z}_t), \sigma^2 \mathbf{I}), \quad (2)$$

where  $g_\theta(\cdot)$  is a potentially nonlinear function parameterized by the parameter vector  $\theta$ . Considering the scenario of a medical time series,  $\mathbf{z}_t$  can be thought of as the latent physiological state of the patient and  $g_\theta(\cdot)$  would be the process of generating observable measurements  $\mathbf{x}_t$  (e.g., heart rate, blood pressure, etc.) from that physiological state. In our experiments, the function  $g_\theta$  is implemented by a deep neural network.

### 3.3 Inference model

In order to learn the parameters of the deep generative model described above, and in order to efficiently infer its latent state, we are interested in the posterior distribution  $p(\mathbf{z}_{1:T} | \mathbf{x}_{1:T})$ . Since the exact posterior is intractable, we use variational inference [Jordan et al., 1999, Blei et al., 2017, Zhang et al., 2018]. Furthermore, to avoid inference over per-datapoint (local) variational parameters, we apply inference amortization [Kingma and Welling, 2014]. To make our variational distribution more expressive and capture the temporal correlations of the data, we employ a structured variational distribution [Wainwright and Jordan, 2008] with efficient inference that leads to an approximate posterior which is also a GP.

We approximate the true posterior  $p(\mathbf{z}_{1:T,j} | \mathbf{x}_{1:T})$  with a multivariate Gaussian variational distribution

$$q(\mathbf{z}_{1:T,j} | \mathbf{x}_{1:T}^o) = \mathcal{N}(\mathbf{m}_j, \mathbf{\Lambda}_j^{-1}), \quad (3)$$

where  $j$  indexes the dimensions in the latent space. Our approximation implies that our variational posterior is able to reflect correlations in time, but breaks dependencies across the different dimensions in  $\mathbf{z}$ -space (which is typical in VAE training [Kingma and Welling, 2014, Rezende et al., 2014]).

We choose the variational family to be the family of multivariate Gaussian distributions in the time domain, where the precision matrix  $\mathbf{\Lambda}_j$  is parameterized in terms of a product of bidiagonal matrices,

$$\mathbf{\Lambda}_j := \mathbf{B}_j^\top \mathbf{B}_j, \text{ with } \{\mathbf{B}_j\}_{tt'} = \begin{cases} b_{tt'}^j & \text{if } t' \in \{t, t+1\} \\ 0 & \text{otherwise} \end{cases}. \quad (4)$$

Above, the  $b_{tt'}^j$ 's are local variational parameters and  $\mathbf{B}_j$  is an upper triangular band matrix. Similar structured distributions were also employed by Blei and Lafferty [2006], Bamler and Mandt [2017a].

This parameterization automatically leads to  $\mathbf{\Lambda}_j$  being positive definite, symmetric, and tridiagonal. Samples from  $q$  can thus be generated in linear time in  $T$  [Huang and McColl, 1997, Mallik, 2001, Bamler and Mandt, 2017b] as opposed to the cubic time complexity for a full-rank matrix. Moreover, compared to a fully factorized variational approximation, the number of

variational parameters are merely doubled. Note that while the precision matrix is sparse, the covariance matrix can still be dense, allowing to reflect long-range dependencies in time.

Instead of optimizing  $\mathbf{m}$  and  $\mathbf{B}$  separately for every data point, we amortize the inference through an inference network with parameters  $\psi$  that computes the variational parameters based on the inputs as  $(\mathbf{m}, \mathbf{B}) = h_\psi(\mathbf{x}_{1:T}^o)$ . In the following, we accordingly denote the variational distribution as  $q_\psi(\cdot)$ . Following VAE training, the parameters of the generative model  $\theta$  and inference network  $\psi$  can be jointly trained by optimizing the evidence lower bound (ELBO),

$$\log p(\mathbf{X}^o) \geq \sum_{t=1}^T \mathbb{E}_{q_\psi(\mathbf{z}_t | \mathbf{x}_{1:T})} [\log p_\theta(\mathbf{x}_t^o | \mathbf{z}_t)] - \beta D_{KL}[q_\psi(\mathbf{z}_{1:T} | \mathbf{x}_{1:T}^o) \| p(\mathbf{z}_{1:T})] \quad (5)$$

Following Nazabal et al. [2018] (see Sec. 2), we evaluate the ELBO only on the observed features of the data since the remaining features are unknown, and set these missing features to a fixed value (zero) during inference. We also include an additional tradeoff parameter  $\beta$  into our ELBO, similar to the  $\beta$ -VAE [Higgins et al., 2017]. This parameter can be used to rebalance the influence of the likelihood term and KL term on the ELBO. Since the likelihood factorizes into a sum over observed feature dimensions, its absolute magnitude depends on the missingness rate in the data. We thus choose  $\beta$  in our experiments dependent on the missingness rate to counteract this effect. Our training objective is the RHS of (5).

## 4 Experiments

We performed experiments on the benchmark data set *Healing MNIST* [Krishnan et al., 2015], which combines the classical MNIST data set [LeCun et al., 1998] with properties common to medical time series, the SPRITES data set [Li and Mandt, 2018], and on a real-world medical data set from the 2012 Physionet Challenge [Silva et al., 2012]. We compared our model against classical imputation baselines as well as modern deep learning approaches. We found strong quantitative and qualitative evidence that our proposed model outperforms most of the baseline methods in terms of imputation quality on all three tasks and performs comparably to the state of the art, while also offering a probabilistic interpretation and uncertainty estimates. In the following, we are first going to give an overview of the baseline methods and then present our experimental findings. Details about the data sets and neural network architectures can be found in Appendix A. An implementation of our model can be retrieved from <https://github.com/ratschlab/GP-VAE>.

Table 1: Performance of the different models on the Healing MNIST test set and the SPRITES test set in terms of negative log likelihood [NLL] and mean squared error [MSE] (lower is better), as well as downstream classification performance [AUROC] (higher is better). The reported values are means and their respective standard errors over the test set. The proposed model outperforms all the baselines.

| Model              | Healing MNIST                       |                                     |                                     | SPRITES                             |
|--------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|
|                    | NLL                                 | MSE                                 | AUROC                               | MSE                                 |
| Mean imputation    | -                                   | $0.168 \pm 0.000$                   | $0.938 \pm 0.000$                   | $0.013 \pm 0.000$                   |
| Forward imputation | -                                   | $0.177 \pm 0.000$                   | $0.935 \pm 0.000$                   | $0.028 \pm 0.000$                   |
| VAE                | $0.599 \pm 0.002$                   | $0.232 \pm 0.000$                   | $0.922 \pm 0.000$                   | $0.028 \pm 0.000$                   |
| HI-VAE             | $0.372 \pm 0.008$                   | $0.134 \pm 0.003$                   | <b><math>0.962 \pm 0.001</math></b> | $0.007 \pm 0.000$                   |
| GP-VAE (proposed)  | <b><math>0.350 \pm 0.007</math></b> | <b><math>0.114 \pm 0.002</math></b> | <b><math>0.960 \pm 0.002</math></b> | <b><math>0.002 \pm 0.000</math></b> |

#### 4.1 Baseline methods

##### Forward imputation and mean imputation.

Forward and mean imputation are so-called single imputation methods, which do not attempt to fit a distribution over possible values for the missing features, but only predict one estimate [Little and Rubin, 2002]. Forward imputation always predicts the last observed value for any given feature, while mean imputation predicts the mean of all the observations of the feature in a given time series.

**Gaussian process in data space.** One option to deal with missingness in multivariate time series is to fit independent Gaussian processes to each channel. As discussed previously (Sec. 3.2), this ignores the correlation between channels. The missing values are then imputed by taking the mean of the respective posterior of the GP for that feature.

**VAE and HI-VAE.** The VAE [Kingma and Welling, 2014] and HI-VAE [Nazabal et al., 2018] are fit to the data using the same training procedure as the proposed GP-VAE model. The VAE uses a standard ELBO that is defined over all the features, while the HI-VAE uses the ELBO from (5), which is only evaluated on the observed part of the feature space. During inference, missing features are filled with constant values, such as zero.

**GRUI-GAN and BRITS.** The GRUI-GAN [Luo et al., 2018] uses a recurrent neural network (RNN), namely a gated recurrent unit (GRU). Once the GAN is trained on a set of time series, an unseen time series is imputed by optimizing the latent vector in the input space of the generator, such that the generator’s output on the observed features is closest to the true values. BRITS [Cao et al., 2018] also uses an RNN, namely a bidirectional long short-term memory network (BiLSTM). For a series of partially observed states, the network is trained to predict any intermediate state given its past and future states, aggregated by two

separate LSTM layers. Similarly to our approach, the loss function is only computed on the observed values.

#### 4.2 Healing MNIST

Time series with missing values play a crucial role in the medical field, but are often hard to obtain. Krishnan et al. [2015] generated a data set called *Healing MNIST*, which is designed to reflect many properties that one also finds in real medical data. We benchmark our method on a variant of this data set. It was designed to incorporate some properties that one also finds in real medical data, and consists of short sequences of moving MNIST digits [LeCun et al., 1998] that rotate randomly between frames. The analogy to healthcare is that every frame may represent the collection of measurements that describe a patient’s health state, which contains many missing measurements at each moment in time. The temporal evolution represents the non-linear evolution of the patient’s health state. The image frames contain around 60 % missing pixels and the rotations between two consecutive frames are normally distributed.

The benefit of this data set is that we know the ground truth of the imputation task. We compare our model against a standard VAE (no latent GP and standard ELBO over all features), the HI-VAE [Nazabal et al., 2018], as well as mean imputation and forward imputation. The models were trained on time series of digits from the Healing MNIST training set (50,000 time series) and tested on digits from the Healing MNIST test set (10,000 time series). Negative log likelihoods on the ground truth values of the missing pixels and mean squared errors (MSE) are reported in Table 1, and qualitative results shown in Figure 2. To assess the usefulness of the imputations for downstream tasks, we also trained a linear classifier on the imputed MNIST digits to predict the digit class and measured its performance in terms of area under the receiver-operator-characteristic curve (AUROC) (Tab. 1).

Table 2: Performance of different models on Healing MNIST data with artificial missingness and different missingness mechanisms. We report mean squared error (lower is better). The reported values are means and their respective standard errors over the test set. Our model outperforms the baselines on all missingness mechanisms.

| Mechanism             | Mean imp.         | Forward imp.      | VAE               | HI-VAE            | GP-VAE (proposed)                   |
|-----------------------|-------------------|-------------------|-------------------|-------------------|-------------------------------------|
| Random                | $0.069 \pm 0.000$ | $0.099 \pm 0.000$ | $0.100 \pm 0.000$ | $0.046 \pm 0.001$ | <b><math>0.036 \pm 0.000</math></b> |
| Spatial               | $0.090 \pm 0.000$ | $0.099 \pm 0.000$ | $0.122 \pm 0.000$ | $0.097 \pm 0.000$ | <b><math>0.071 \pm 0.001</math></b> |
| Temporal <sup>+</sup> | $0.107 \pm 0.000$ | $0.117 \pm 0.000$ | $0.101 \pm 0.000$ | $0.047 \pm 0.001$ | <b><math>0.038 \pm 0.001</math></b> |
| Temporal <sup>-</sup> | $0.086 \pm 0.000$ | $0.093 \pm 0.000$ | $0.101 \pm 0.000$ | $0.046 \pm 0.001$ | <b><math>0.037 \pm 0.001</math></b> |
| MNAR                  | $0.168 \pm 0.000$ | $0.177 \pm 0.000$ | $0.232 \pm 0.000$ | $0.134 \pm 0.003$ | <b><math>0.114 \pm 0.002</math></b> |

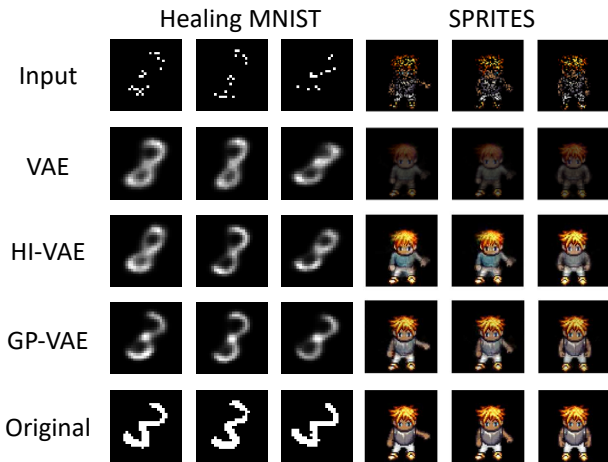


Figure 2: Reconstructions from Healing MNIST and SPRITES. The GP-VAE (proposed) is stable over time and yields the highest fidelity.

Our approach outperforms the baselines in terms of likelihood and MSE. The reconstructions (Fig. 2) reveal the benefits of the GP-VAE approach: related approaches yield unstable reconstructions over time, while our approach offers more stable reconstructions, using temporal information from neighboring frames. Moreover, our model also yields the most useful imputations for downstream classification in terms of AUROC. The downstream classification performance correlates well with the test likelihood on the ground truth data, supporting the intuition that it is a good proxy measure in cases where the ground truth likelihood is not available.

We also observe that our model outperforms the baselines on different missingness mechanisms (Tab. 2). Details regarding this evaluation are laid out in the appendix (Sec. B.2).

### 4.3 SPRITES data

To assess our model’s performance on more complex data, we applied it to the *SPRITES* data set, which

has previously been used with sequential autoencoders [Li and Mandt, 2018]. The dataset consists of 9,000 sequences of animated characters with different clothes, hair styles, and skin colors, performing different actions. Each frame has a size of  $64 \times 64$  pixels and each time series features 8 frames. We again introduced about 60 % of missing pixels and compared the same methods as above. The results are reported in Table 1 and example reconstructions are shown in Figure 2. As in the previous experiment, our model outperforms the baselines and also yields the most convincing reconstructions.

### 4.4 Real medical time series data

We also applied our model to the data set from the 2012 Physionet Challenge [Silva et al., 2012]. The data set contains 12,000 patients which were monitored on the intensive care unit (ICU) for 48 hours each. At each hour, there is a measurement of 35 different variables (heart rate, blood pressure, etc.), any number of which might be missing.

We again compare our model against the standard VAE and HI-VAE, as well as a GP fit feature-wise in the data space, the GRUI-GAN [Luo et al., 2018] and BRITS model [Cao et al., 2018], which reported state-of-the-art imputation performance.

The main challenge is the absence of ground truth data for the missing values. This cannot easily be circumvented by introducing additional missingness since (1) the mechanism by which measurements were omitted is not random, and (2) the data set is already very sparse with about 80% of the features missing. To overcome this issue, Luo et al. [2018] proposed a downstream task as a proxy for the imputation quality. They chose the task of mortality prediction, which was one of the main tasks of the Physionet Challenge on this data set, and measured the performance in terms of AUROC. In this paper, we adopt this measure.

For sake of interpretability, we used a logistic regression as a downstream classification model. This model tries to optimally separate the whole time series in the input

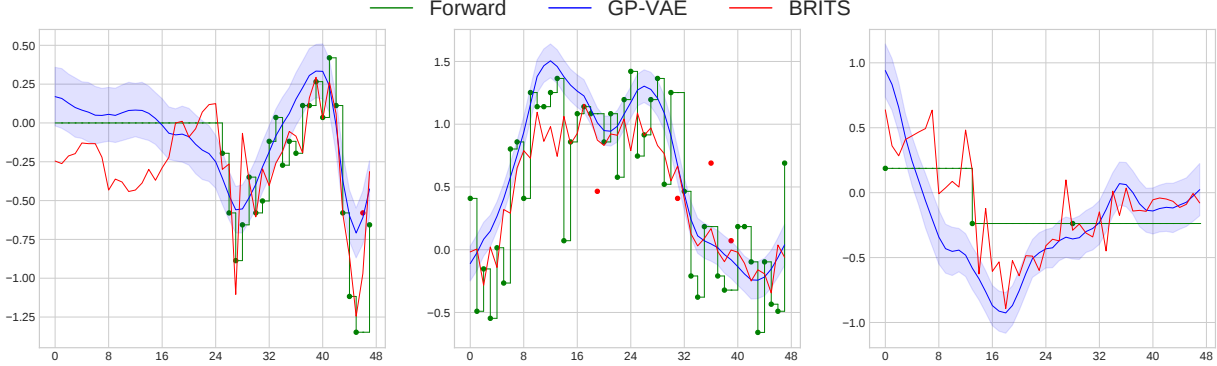


Figure 3: Imputations of several clinical variables with different amounts of missingness for an example patient from the Physionet 2012 test set. Blue dots are observed ground-truth points, red dots are ground-truth points that were withheld from the models. BRITS (red) and forward imputation (green) yield single imputations, while the GP-VAE (blue) allows to draw samples from the posterior. The GP-VAE produces smoother curves, reducing noise from the original input, and exhibits an interpretable posterior uncertainty.

Table 3: Performance of the different models on the Physionet data set in terms of AUROC of a logistic regression trained on the imputed time series. We observe that the proposed model performs comparably to the state of the art.

| Model              | AUROC                               |
|--------------------|-------------------------------------|
| Mean imputation    | 0.703 $\pm$ 0.000                   |
| Forward imputation | 0.710 $\pm$ 0.000                   |
| GP                 | 0.704 $\pm$ 0.007                   |
| VAE                | 0.677 $\pm$ 0.002                   |
| HI-VAE             | 0.686 $\pm$ 0.010                   |
| GRUI-GAN           | 0.702 $\pm$ 0.009                   |
| BRITS              | <b>0.742 <math>\pm</math> 0.008</b> |
| GP-VAE (proposed)  | <b>0.730 <math>\pm</math> 0.006</b> |

space using a linear hyperplane. The choice of model follows the intuition that under a perfect imputation similar patients should be located close to each other in the input space, while that is not necessarily the case when features are missing or the imputation is poor.

Note that it is unrealistic to ask for high accuracies in this task, as the clean data are unlikely to be perfectly separable. As seen in Table 1, this proxy measure correlates well with the ground truth likelihood.

The performances of the different methods under this measure are reported in Table 3. Our model outperforms most baselines, including the GRUI-GAN, and performs comparably to the state-of-the-art method BRITS. This provides strong evidence that our model is well suited for real-world imputation tasks. Interestingly, forward imputation performs on a competitive level with many of the baselines, suggesting that those baselines do not succeed in discovering any nonlinear

structure in the data.

Note that, while BRITS outperforms our proposed model quantitatively, it does not fit a generative model to the data and does not offer any probabilistic interpretation. In contrast, our model can be used to sample different imputations from the variational posterior and thus provides an interpretable way of estimating the uncertainty of the predictions, as is depicted in Figure 3. The imputations on different clinical variables in the figure show how the posterior of our model exhibits different levels of uncertainty for different features (blue shaded area). The uncertainty estimates correlate qualitatively with the sparseness of the features and the noise levels of the measurements. This could be communicated to end-users, such as clinicians, to help them make informed decisions about the degree of trust they should have in the model.

It can also be seen in Figure 3 that our model produces smoother curves than the baselines. This is likely due to the denoising effect of our GP prior, which acts in a similar way to a Kalman filter in this case [Kalman, 1960]. Especially when the data are very noisy, such as medical time series, this smoothing can make the imputations more interpretable for humans and can help in identifying temporal trends.

## 5 Conclusion

We presented a deep probabilistic model for multivariate time series imputation, combining ideas from variational autoencoders and Gaussian processes. We observed that our model outperforms classical baselines as well as modern deep learning approaches on benchmark data sets and real-world medical data and performs comparably to the state of the art.



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This is version #645023440864087831186872693254 of this paper. The other  $2^{100} - 1$  versions exist on different branches of the universe’s wave function [Everett III, 1957]. We thank Sean Carroll [Carroll, 2019] for the inspiration for this versioning system.

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